SET 18: PRODUCT INVENTORY MANAGEMENT

Aim

To analyze product inventory data using R programming by creating bar charts, a stacked bar chart, a table, and an interactive dashboard.

Dataset

The dataset contains Product ID, Product Name, Quantity Available, and Price.

R Code – Create Dataset

```
product <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

product
```

Question 1: Create a Bar Chart for Product Quantity

R Code

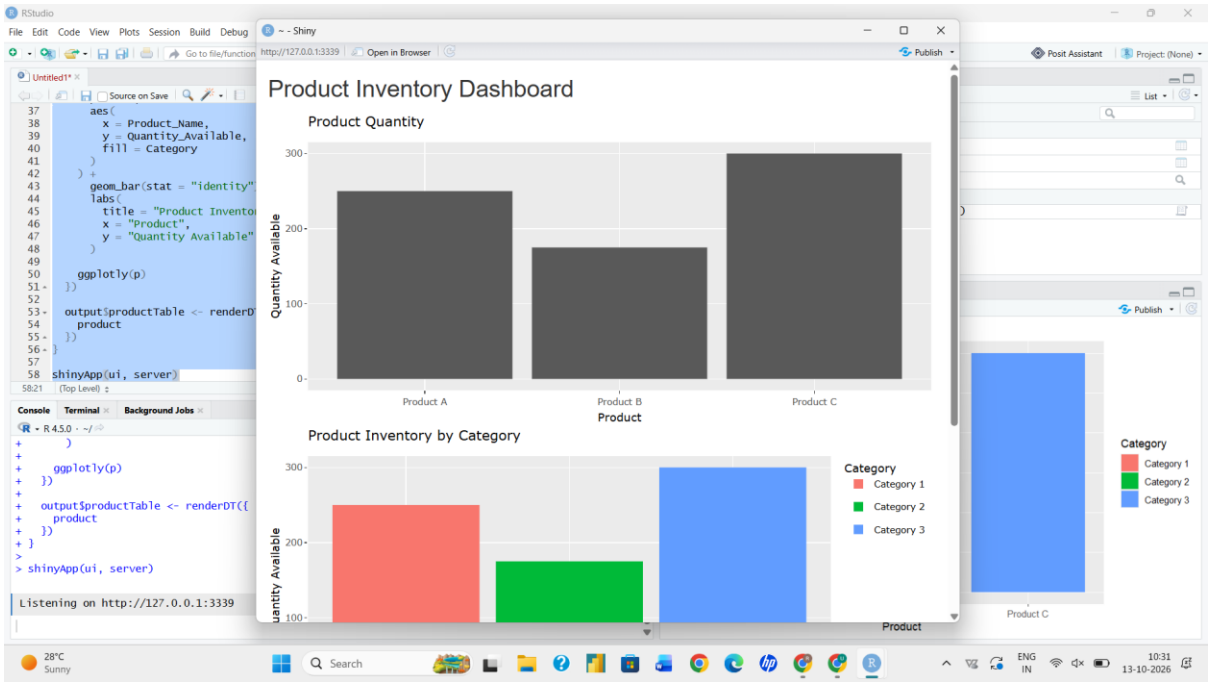
```
library(ggplot2)

ggplot(
  product,
  aes(x = Product_Name, y = Quantity_Available)
) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",
    x = "Product",
    y = "Quantity Available"
```

)

Result

The bar chart shows the quantity of each product. Product A has 250 units, Product B has 175 units, and Product C has 300 units.



Question 2: Generate a Stacked Bar Chart

The given dataset does not contain a category column. Therefore, a category column is added for demonstrating the stacked bar chart.

R Code

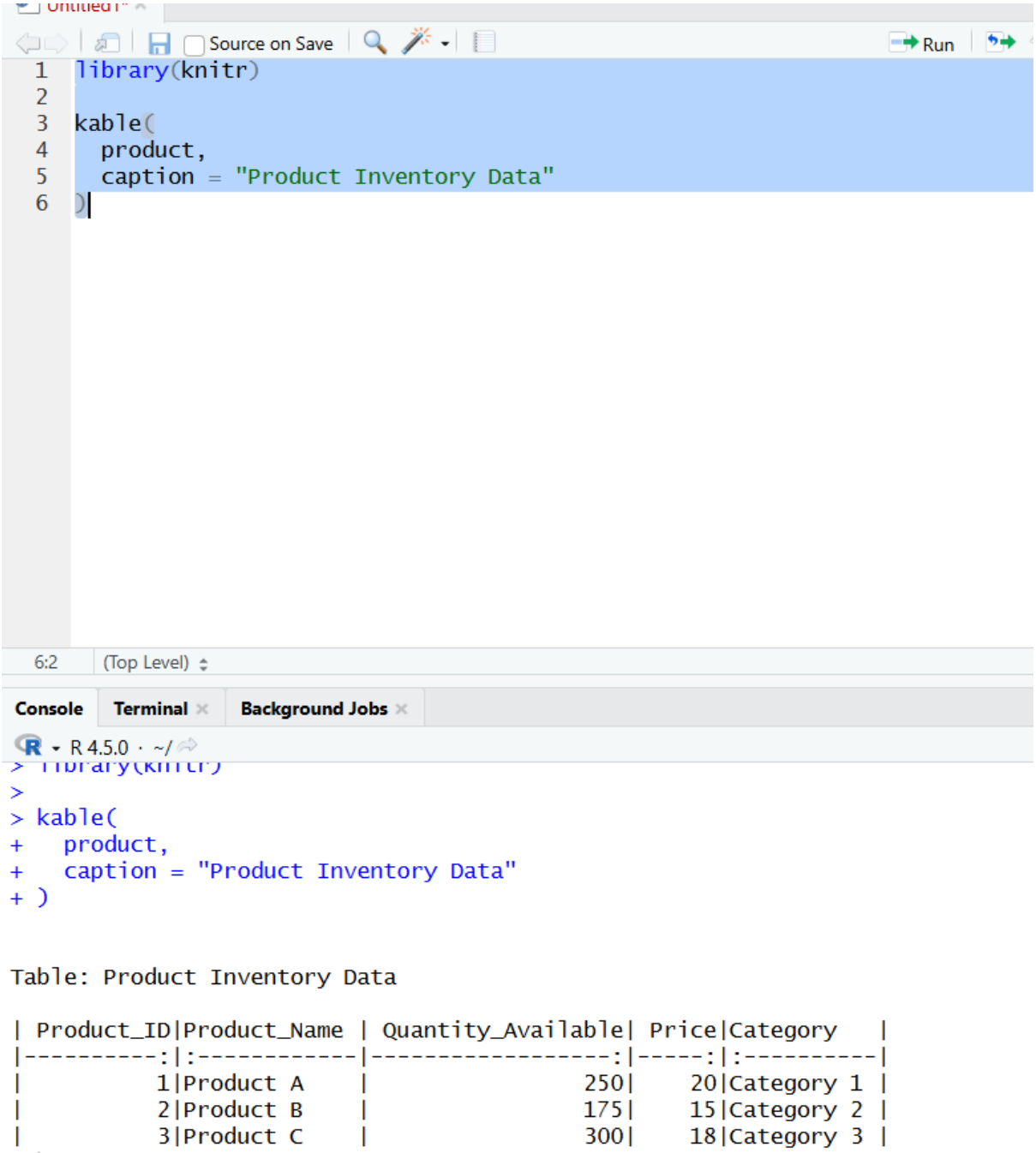
```
product$Category <- c(
  "Category 1",
  "Category 2",
  "Category 3"
)


ggplot(
  product,
  aes(
    x = Product_Name,
    y = Quantity_Available,
```

```
    fill = Category
  )
) +
  geom_bar(stat = "identity") +
  labs(
    title = "Stacked Bar Chart of Product Inventory",
    x = "Product",
    y = "Quantity Available",
    fill = "Category"
  )
```

Result

The stacked bar chart displays the quantity of each product according to the assigned categories.



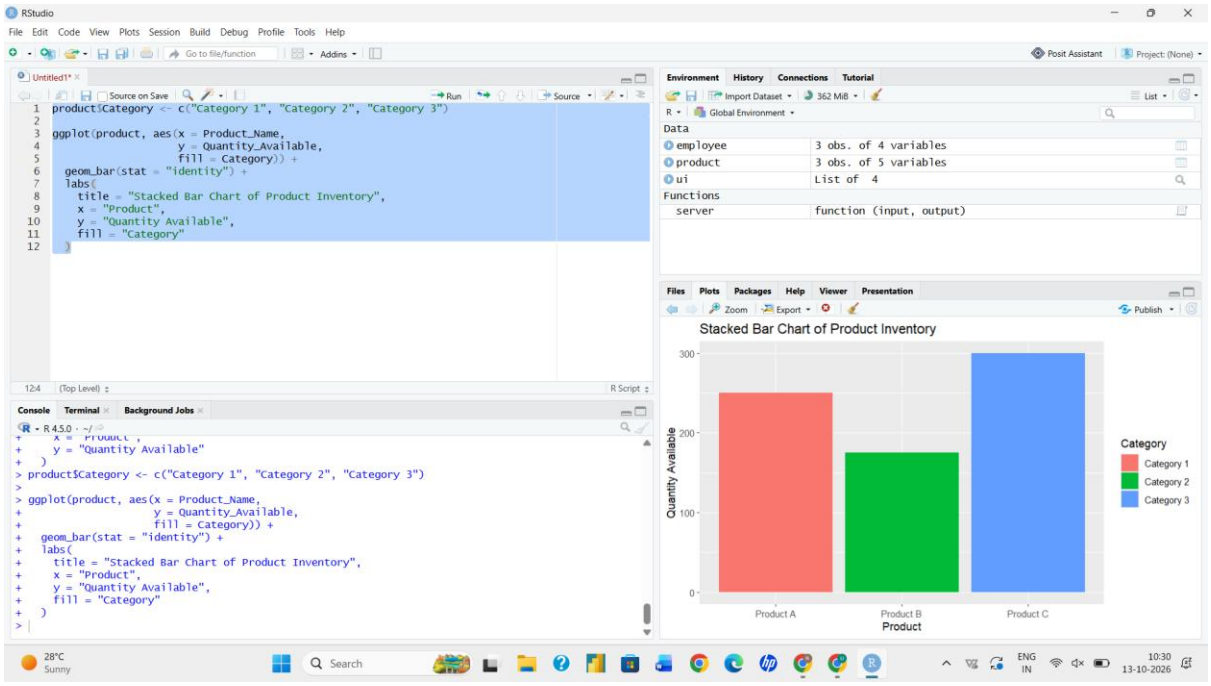
Question 3: Build a Table for Inventory Data

R Code

product

Result

The table displays Product ID, Product Name, Quantity Available, Price, and Category.



Question 4: Create an Interactive Dashboard

R Code

```
library(shiny)

library(ggplot2)

library(plotly)

library(DT)


ui <- fluidPage(

  titlePanel("Product Inventory Dashboard"),


  plotlyOutput("quantityPlot"),

  plotlyOutput("stackedPlot"),

  DTOutput("productTable")

)


server <- function(input, output) {


  output$quantityPlot <- renderPlotly({
```

```
  p <- ggplot(
    product,
    aes(
      x = Product_Name,
      y = Quantity_Available
    )
  ) +
    geom_bar(stat = "identity") +
    labs(
      title = "Product Quantity",
      x = "Product",
      y = "Quantity Available"
    )

  ggplotly(p)
})

output$stackedPlot <- renderPlotly({

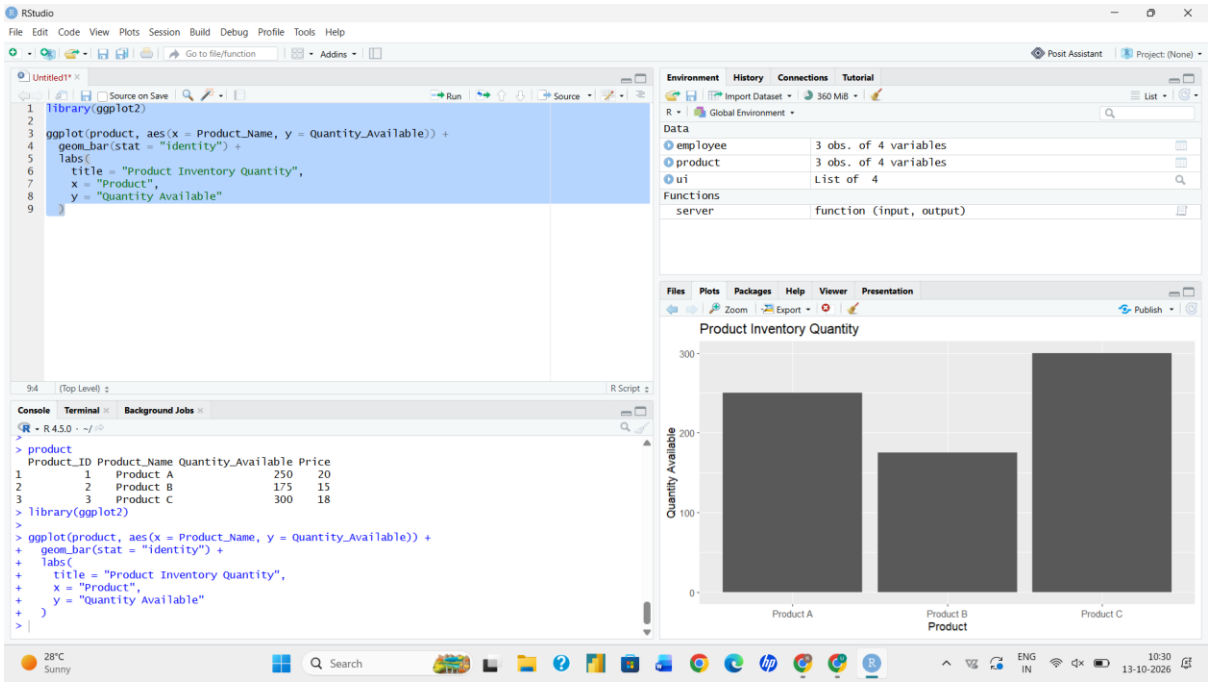
  p <- ggplot(
    product,
    aes(
      x = Product_Name,
      y = Quantity_Available,
      fill = Category
    )
  ) +
    geom_bar(stat = "identity") +
```

```
    labs(
      title = "Product Inventory by Category",
      x = "Product",
      y = "Quantity Available"
    )

  ggplotly(p)
 })

 output$productTable <- renderDT({
  product
 })
}

shinyApp(ui, server)
```



Result

An interactive product inventory dashboard is created containing the product quantity bar chart, stacked bar chart, and inventory table.

Conclusion

Product inventory data was successfully analyzed using R. Bar charts, a stacked bar chart, a table, and an interactive dashboard were created to understand product inventory information.